

Supplementary Data

lifelines-hc: a Python package for Higher Criticism testing in two-sample survival analysis under non-proportional hazards

Alon Kipnis, Ben Galili, Zohar Yakhini

This document accompanies the Application Note *lifelines-hc: a Python package for Higher Criticism testing in two-sample survival analysis under non-proportional hazards*. It provides (i) the full analysis workflow and scripts used to reproduce the case studies, and (ii) complete per-method test statistics and p -values for all four datasets analysed in the main text.

1 Analysis scripts and data

All scripts live in the `AppNote/` directory of the `lifelines-hc` repository (<https://github.com/alonkipnis/lifelines-hc/tree/main/AppNote>). They require Python ≥ 3.8 , `lifelines-hc`, and `lifelines` ≥ 0.29 . The MaxCombo and Yang–Prentice comparisons additionally require R with the CRAN packages `nph` [Ristl et al., 2021] and `YPmodel` [Yang and Prentice, 2005, 2010].

Workflow

1. **Download trial IPD.** `01_get_kmdata.py` exports reconstructed individual patient data (IPD) from the `kmdata` R package [Guyot et al., 2012] for CheckMate 057 [Borghaei et al., 2015], COMET-1 [Smith et al., 2016], and AZURE [Coleman et al., 2011, 2014]. CSL liver-cirrhosis IPD are downloaded directly from the CRAN `timereg` package and the `SurvSet` repository [Martinussen and Scheike, 2006, Drysdale, 2022] (see `run_csl.py`).
2. **Run per-dataset analyses.** Each `run_*.py` script loads one dataset, calls `run_all_tests()` in `utils.py`, writes a results CSV to `results/`, and saves diagnostic figures to `figs/`. `run_all_tests()` also invokes `competitors.R` for MaxCombo and Yang–Prentice.
3. **Build the composite figure.** `make_figure.py` assembles Figure 1 of the main text from the four domain scripts.

Script index

Reproduction commands

```
cd AppNote
python 01_get_kmdata.py --trials Checkmate057_1C COMET1_2A AZURE_2A
python run_immuno_oncology.py
python run_comet.py
python run_azure.py
python run_csl.py           # requires SurvSet or AppNote/data/CSL.csv
python make_figure.py
```

Table 1: Analysis scripts for the four case studies.

Script	Dataset	Role
<code>01_get_kmdata.py</code>	CheckMate 057, COMET-1, AZURE	Download IPD via <code>kmdata</code> (Guyot algorithm).
<code>run_immuno_oncology.py</code>	CheckMate 057 PFS ($n = 582$)	Nivolumab vs. docetaxel; crossing curves.
<code>run_comet.py</code>	COMET-1 OS ($n = 1028$)	Cabozantinib vs. prednisone; transient early benefit.
<code>run_azure.py</code>	AZURE DFS ($n = 3359$)	Zoledronic acid vs. control; menopause-dependent effect.
<code>run_csl.py</code>	CSL OS ($n = 446$)	Prednisone vs. placebo; public raw IPD, multi-window effect.
<code>utils.py</code>	—	Shared <code>run_all_tests()</code> , plotting, and table utilities.
<code>competitors.R</code>	—	MaxCombo (<code>nph::logrank.maxtest</code>) and Yang-Prentice adaptive log-rank (<code>YPmodel::YPmodel.adlgrk</code>).
<code>make_figure.py</code>	All four	Composite Figure 1 panels (A–H).

```
# MaxCombo / Yang--Prentice: requires R packages nph and YPmodel
#   install.packages(c("nph", "YPmodel"))
# invoked automatically from run_all_tests() via competitors.R
```

Each `run_*.py` script writes `results/<domain>_test_results.csv` containing one row per competing method (statistic and permutation- or asymptotic-based p -value).

2 Data sources

Table 2 lists the original clinical studies, endpoints, and how each dataset was obtained. For CheckMate 057, COMET-1, and AZURE, only published Kaplan–Meier curves were available; IPD were reverse-engineered with the algorithm of Guyot et al. [2012] and distributed in the `kmdata` package (<https://github.com/raredd/kmdata>). The CSL trial is the only case study with genuinely public raw IPD.

Reconstructed IPD files are written to `AppNote/data/` as `Checkmate057_1C.csv`, `COMET1_2A.csv`, and `AZURE_2A.csv` by `01_get_kmdata.py`. CSL data are written to `AppNote/data/CSL.csv` by `run_csl.py`.

3 Shared analysis settings

Unless noted otherwise, HCHG uses:

- equal-width time pooling with `n_intervals_to_pool` as listed per dataset below;
- two-sided interval hypergeometric p -values (`alternative='both'`);
- HC fraction $\gamma = 0.2$;
- label-permutation p -values with the random seed fixed at 42.

Table 2: Original studies and data acquisition for the four case studies.

Trial	Endpoint	n	Source and acquisition
CheckMate 057	PFS	582	Borghaei et al. [2015] ; IPD reconstructed from published Kaplan–Meier curves via Guyot et al. [2012] (<code>kmdata</code>).
COMET-1	OS	1028	Smith et al. [2016] ; IPD reconstructed from published Kaplan–Meier curves via Guyot et al. [2012] (<code>kmdata</code>).
AZURE	DFS	3359	Coleman et al. [2011, 2014] ; IPD reconstructed from published Kaplan–Meier curves via Guyot et al. [2012] (<code>kmdata</code>).
CSL	OS	446	Copenhagen liver-cirrhosis trial (1962–1969); raw IPD from the CRAN <code>timereg</code> package [Martinussen and Scheike, 2006] and <code>SurvSet</code> [Drysdale, 2022].

The log-rank test and four weighted log-rank variants (Gehan-Wilcoxon, Tarone-Ware, Peto-Prentice, Fleming-Harrington with $p = q = 1$) use the standard `lifelines.statistics.logrank_test` implementation with asymptotic p -values.

MaxCombo is the two-sided maximum of the Fleming-Harrington weighted log-rank Z -statistics with $(\rho, \gamma) \in \{(0, 0), (0, 1), (1, 0), (1, 1)\}$ — the four-weight immuno-oncology dictionary of [Lin et al. \[2020\]](#) — with p -value from the estimated multivariate normal joint distribution as implemented in `nph::logrank.maxtest` [[Ristl et al., 2021](#)] (`alternative='two.sided'`, random seed 42). The reported statistic is $z_{\max} = \max_j |Z_j|$. The Yang-Prentice adaptive weighted log-rank test of [Yang and Prentice \[2010\]](#), based on the short-term/long-term hazard-ratio model of [Yang and Prentice \[2005\]](#), is computed via `YPmodel::YPmodel.adlgrk`; the reported statistic is the maximum of the two adaptive $|Z|$ components used internally by that procedure.

4 Per-method results tables

Tables 3–6 report the full comparison for each dataset. Asterisks mark $p < 0.05$. Permutation-based p -values for HCHG are discrete (multiples of $1/(n_{\text{perms}} + 1)$ with $n_{\text{perms}} = 2000$) and are rounded to three decimals; “ < 0.001 ” denotes the smallest attainable value, $1/2001$.

CheckMate 057 — progression-free survival ($n = 582$)

Original trial: [Borghaei et al. \[2015\]](#). IPD reconstructed from published Kaplan–Meier curves [[Guyot et al., 2012](#)]. Control: docetaxel ($n = 290$); treatment: nivolumab ($n = 292$). HC settings: $K = 60$ intervals, $n_{\text{perms}} = 2000$.

MaxCombo is driven by the late-weighted FH(0, 1) member ($Z = 4.27$, $p = 2.0 \times 10^{-5}$); FH(0, 0), FH(1, 0), and FH(1, 1) have two-sided $p = 0.351$, 0.058, and 0.256. The Yang-Prentice fit yields short-term hazard ratio $\hat{\theta}_1 = 2.33$ and long-term ratio $\hat{\theta}_2 = 0.44$.

Table 3: CheckMate 057 PFS: all competing methods.

Method	Statistic	p -value
Log-rank	0.868	0.351
Gehan-Wilcoxon	4.169	0.041*
Tarone-Ware	0.844	0.358
Peto-Prentice	3.869	0.049*
Fleming-Harrington (1,1)	1.292	0.256
MaxCombo	4.268	2.5×10^{-5} *
Yang-Prentice	3.626	0.00055*
HCHG	1.695	0.001*

COMET-1 — overall survival ($n = 1028$)

Original trial: [Smith et al. \[2016\]](#). IPD reconstructed from published Kaplan–Meier curves [[Guyot et al., 2012](#)]. Control: prednisone ($n = 346$); treatment: cabozantinib ($n = 682$). HC settings: $K = 80$ intervals, $n_{\text{perms}} = 2000$.

Table 4: COMET-1 OS: all competing methods.

Method	Statistic	p -value
Log-rank	1.259	0.262
Gehan-Wilcoxon	1.693	0.193
Tarone-Ware	1.570	0.210
Peto-Prentice	1.602	0.206
Fleming-Harrington (1,1)	0.648	0.421
MaxCombo	1.258	0.338
Yang-Prentice	1.176	0.262
HCHG	1.179	0.014*

AZURE — disease-free survival ($n = 3359$)

Original trial: [Coleman et al. \[2011, 2014\]](#). IPD reconstructed from published Kaplan–Meier curves [[Guyot et al., 2012](#)]. Control: standard adjuvant therapy ($n = 1678$); treatment: zoledronic acid ($n = 1681$). HC settings: $K = 80$ intervals, $n_{\text{perms}} = 2000$.

CSL — overall survival ($n = 446$)

Original trial: Copenhagen Study Group for Liver Diseases (1962–1969); public IPD distributed via [Martinussen and Scheike \[2006\]](#) and [Drysdale \[2022\]](#). Control: placebo ($n = 220$); treatment: prednisone ($n = 226$). Follow-up times are in years. HC settings: $K = 50$ intervals, $n_{\text{perms}} = 2000$.

5 Cross-dataset summary

Table 7 condenses the log-rank, MaxCombo, Yang–Prentice, and HCHG p -values reported in the main text. Full statistics for all eight methods appear in Section 4.

Table 5: AZURE DFS: all competing methods.

Method	Statistic	p -value
Log-rank	1.054	0.305
Gehan-Wilcoxon	0.652	0.419
Tarone-Ware	0.691	0.406
Peto-Prentice	1.145	0.284
Fleming-Harrington (1,1)	0.277	0.598
MaxCombo	1.072	0.401
Yang-Prentice	1.044	0.308
HCHG	1.513	0.005*

Table 6: CSL liver-cirrhosis OS: all competing methods.

Method	Statistic	p -value
Log-rank	0.757	0.384
Gehan-Wilcoxon	0.439	0.507
Tarone-Ware	0.774	0.379
Peto-Prentice	0.574	0.449
Fleming-Harrington (1,1)	2.234	0.135
MaxCombo	1.495	0.236
Yang-Prentice	0.872	0.393
HCHG	1.257	0.002*

Across all four datasets, HCHG achieves $p \leq 0.014$ while the log-rank test is non-significant throughout ($p \geq 0.26$). MaxCombo and Yang-Prentice detect the delayed-benefit crossing on CheckMate 057 but are non-significant on COMET-1, AZURE, and CSL ($p \geq 0.24$). Among the single weighted log-rank alternatives, only CheckMate 057 Gehan-Wilcoxon and Peto-Prentice reach $p < 0.05$ (at 0.041 and 0.049 respectively).

6 Machine-readable results

The CSV files below contain the same values as Tables 3–6 (columns `method`, `statistic`, `p-value`):

- AppNote/results/immuno_test_results.csv
- AppNote/results/comet_test_results.csv
- AppNote/results/azure_test_results.csv
- AppNote/results/csl_test_results.csv
- AppNote/results/maxcombo_yp_details.json (MaxCombo component Z -statistics and Yang-Prentice $\hat{\theta}_1, \hat{\theta}_2$).

Table 7: Summary of log-rank, MaxCombo, Yang–Prentice, and HCHG p -values across the four case studies (full tables in Section 4).

Dataset	Endpoint	n	Log-rank	MaxCombo	Yang–Prentice	HCHG
CheckMate 057	PFS	582	0.351	2.5×10^{-5}	0.00055	0.001
COMET-1	OS	1028	0.262	0.338	0.262	0.014
AZURE	DFS	3359	0.305	0.401	0.308	0.005
CSL	OS	446	0.384	0.236	0.393	0.002

References

- Hossein Borghaei, Luis Paz-Ares, Leora Horn, David R. Spigel, Martin Steins, Neal E. Ready, Laura Q. Chow, Everett E. Vokes, Enriqueta Felip, Esther Holgado, et al. Nivolumab versus docetaxel in advanced nonsquamous non–small-cell lung cancer. *New England Journal of Medicine*, 373(17):1627–1639, 2015. doi: 10.1056/NEJMoa1507643.
- Robert E. Coleman, Howard Marshall, David Cameron, David Dodwell, Robert Burkinshaw, Mary Keane, Michael Gil, Stephen J. Houston, Robert J. Grieve, Peter J. Barrett-Lee, et al. Breast-cancer adjuvant therapy with zoledronic acid. *New England Journal of Medicine*, 365(15): 1396–1405, 2011. doi: 10.1056/NEJMoa1105195.
- Robert E. Coleman, Marie Collinson, Eleanor Banks, Graham Clack, Dorothy Ritchie, Victoria Liversedge, David Cameron, David Dodwell, Michael Jones, Peter Barrett-Lee, et al. Benefits and risks of adjuvant treatment with zoledronic acid in stage II/III breast cancer. 10 year follow-up of the AZURE randomized clinical trial. *Journal of Bone Oncology*, 3(2):23–30, 2014. doi: 10.1016/j.jbo.2013.12.001.
- Erik Drysdale. SurvSet: An open-source time-to-event dataset repository. *arXiv preprint arXiv:2203.03094*, 2022. doi: 10.48550/arXiv.2203.03094.
- Patricia Guyot, A. E. Ades, Mario J. N. M. Ouwens, and Nicky J. Welton. Enhanced secondary analysis of survival data: reconstructing the data from published Kaplan-Meier survival curves. *BMC Medical Research Methodology*, 12:9, 2012. doi: 10.1186/1471-2288-12-9.
- Ray S. Lin, Ji Lin, Satrajit Roychoudhury, Keaven M. Anderson, Tianle Hu, Bo Huang, Larry F. Leon, Jason J. Z. Liao, Rong Liu, Xiaodong Luo, Pralay Mukhopadhyay, Rui Qin, Kay Tatsuoka, Xuejing Wang, Yang Wang, Jian Zhu, Tai-Tsang Chen, Renee Iacona, and Cross-Pharma Non-proportional Hazards Working Group. Alternative analysis methods for time to event endpoints under nonproportional hazards: a comparative analysis. *Statistics in Biopharmaceutical Research*, 12(2):187–198, 2020. doi: 10.1080/19466315.2019.1697738.
- Torben Martinussen and Thomas H. Scheike. *Dynamic Regression Models for Survival Data*. Springer, New York, 2006. doi: 10.1007/0-387-33960-4. CSL liver cirrhosis trial (prednisone vs placebo, Copenhagen Study group for Liver diseases, 1962–1969); dataset `csl` in the CRAN `timereg` package.
- Robin Ristl, Nicolas Ballarini, Heiko Götte, Armin Schöler, Martin Posch, and Franz König. Delayed treatment effects, treatment switching and heterogeneous patient populations: how to design and analyze RCTs in oncology. *Pharmaceutical Statistics*, 20(1):129–145, 2021. doi: 10.1002/pst.2062.

Matthew R. Smith, Christopher J. Sweeney, Paul G. Corn, Dana E. Rathkopf, David C. Smith, Maha Hussain, Daniel J. George, Celestia S. Higano, Andrea L. Harzstark, A. Oliver Sartor, et al. Cabozantinib in chemotherapy-pretreated metastatic castration-resistant prostate cancer: results of a phase III randomized controlled trial (COMET-1). *Journal of Clinical Oncology*, 34(25): 3005–3013, 2016. doi: 10.1200/JCO.2015.65.5597.

Song Yang and Ross Prentice. Semiparametric analysis of short-term and long-term hazard ratios with two-sample survival data. *Biometrika*, 92(1):1–17, 2005. doi: 10.1093/biomet/92.1.1.

Song Yang and Ross Prentice. Improved logrank-type tests for survival data using adaptive weights. *Biometrics*, 66(1):30–38, 2010.